

Shritan Kommareddy

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EDUCATION

South Forsyth High School

Cumming, GA

Expected Graduation: 2028 | GPA: 4.00 Unweighted, 4.44 Weighted | Class Rank: 17/728

Dual Enrollment: Georgia State University | May 2026 – Present

Relevant Coursework: AP Computer Science A, AP Precalculus, AP Biology, AP Chemistry, AP Physics 1

EXPERIENCE & LEADERSHIP

Trailblazers Robotics (Reference: (770) 990-4521)

Cumming, GA

Manager & CTO, May 2025 – Present | Robotics Teacher, Summer 2023 – May 2025

- Promoted from Robotics Teacher to Manager & CTO; manage finances, enrollment, parent communication, and coach/tutor operations for a 250-student, six-figure robotics education business.
- Designed an ESP32-based PBL curriculum used by 100+ students; led county- and state-level VEX competitions attended by 500+ people, coordinating volunteers, logistics, and event operations.

Technology Student Association – Member & Sergeant at Arms

Officer since May 2025

- Third-year member competing in robotics, controls, and rigorous CAD/modeling events; researched PID and applied it to design a two-day controls curriculum and simulator, designed custom competition robots, and review 30+ projects across 7+ engineering events.

FIRST Robotics Competition Team 4112 – Design/CAD

Subsystem Lead, 2025–2026

- Led collaborative design, machining, and tuning of the Spinning Indexer, routing balls from intake to shooter; fabricated polycarbonate and aluminum parts, coordinated interfaces with multiple subsystems, and directly supported on-field scoring/control and state-qualifying performance.

Varsity Tennis – South Forsyth High School | Three-year varsity athlete; state semifinalist

2024 – Present

ENGINEERING PROJECTS

StablStep – Multi-Axis Self-Stabilizing Prosthetic Foot

May 2025 – Aug. 2026

- Independently designed and fabricated a full-scale prototype using Arduino, PID, and FDM 3D printing, with modified servo feedback for active multi-axis stabilization.
- Developed a compliant series-elastic-inspired spring mechanism; collaborated with ABC-certified prosthetic technicians at Hanger Clinic on mounting, usability, and universal design.

Open-Source 7-DOF Humanoid Arms

May 2026 – Sept. 2026

- Designed low-cost 7-DOF humanoid arms using Onshape and FDM printing; published the full open-source platform and received project funding through Hack Club Macondo.
- Built a ROS 2 stack with Gazebo, MoveIt, PickIK, Linux, and VR teleoperation; continuing VLA-based work for generalized manipulation tasks.

Assistive Wheelchair Robotic Arm

June 2026 – Present

- Designing a low-cost 6-DOF SCARA-inspired arm for powered-wheelchair users, targeting a 1 lb payload, ~3.5 ft reach, ~5 ft maximum ground-to-end-effector height, compact folding, and backdrivable rotary joints.
- Developing a custom ball-screw lift with aluminum extrusion/linear guides, NEMA 23 stepper, RS02 actuators, bus servos, and TPU gripper; planning ROS 2/CAN control with an accessible joystick control interface.

HONORS & AWARDS

TSA Robotics – National Champion

2026

- Placed 1st nationally among ~150 state-qualified teams; designed and built a fully custom, all-purpose competition robot as part of a four-person team.

The Hive Summer Buildathon – 1st Place

2026

- Won 1st among 125+ international individual submissions and a \$1,000 prize for the Open-Source 7-DOF Humanoid Arms.

DreamHacks Hackathon – Best Overall Submission

2026

- Won Best Overall Submission / 1st Place among 80 teams for TideWater, an end-to-end 311-style community issue reporting platform, alongside a custom vision solution for obstruction detection.

TECHNICAL SKILLS

CAD & Engineering: Onshape, SolidWorks, ArchiCAD, FEA, Mechanical Prototyping

Robotics & Controls: ROS 2, Gazebo, MoveIt, PickIK, PID, CAN Bus | **Programming:** Java, JavaScript, C#

Embedded Systems: Arduino, ESP32 | **Manufacturing:** CNC Machining, FDM 3D Printing, Resin 3D Printing, Soldering

Developer Tools: Docker, Linux